NeuroLens

Watch a brain watch your video.

PREDICTED BRAIN RESPONSE TO VIDEO, SECOND BY SECOND

CS5224 CLOUD COMPUTING · FINAL PRESENTATION · DYLAN · XIN WEI · KELVIN · JOSHUA · BRANDON

$10k+

to test one ad on a brain scanner

fMRI study	$10,000+	days to weeks
Facial coding	$850 – $4,000	hours to days
NeuroLens	$1.00	about 7 min

Cost to evaluate one video. fMRI vs surveys: Venkatraman et al., 2015. Competitor ranges: our preliminary report.
NeuroLens: planned price per 30 s clip.

Ten cuts by lunch.
No way to judge them.

Generative tools made production cheap. The feed made every second count.



One upload in.
A brain response out.

Your video
MP4, MOV or WebM, up to two minutes

TRIBE v2
Meta's brain model, trained on real fMRI of people watching film

20,484 points
of the cortex, predicted every second

Three outputs
Timeline · 3D brain · AI report

For every second of the video. A virtual pre-test of a typical viewer, not a prediction of views or sales.

Two people, two questions.

“Which three seconds do I cut?”

A short-form creator uploads two edits of an intro, finds the dip, and re-cuts.

“Which variant gets the media budget?”

A small agency uploads up to five cuts, compares any pair, and decides with evidence.

Seven brain areas.
Three questions.

Seeing

faces · bodies · scenes · visual

Hearing

sound

Understanding

speech · social

BELOW TYPICAL

0 = TYPICAL FILM & TV VIEWING

ABOVE TYPICAL

From file to insight in five steps.

01
Sign in
Google account

02
Upload
Price shown first, 15 s to cancel

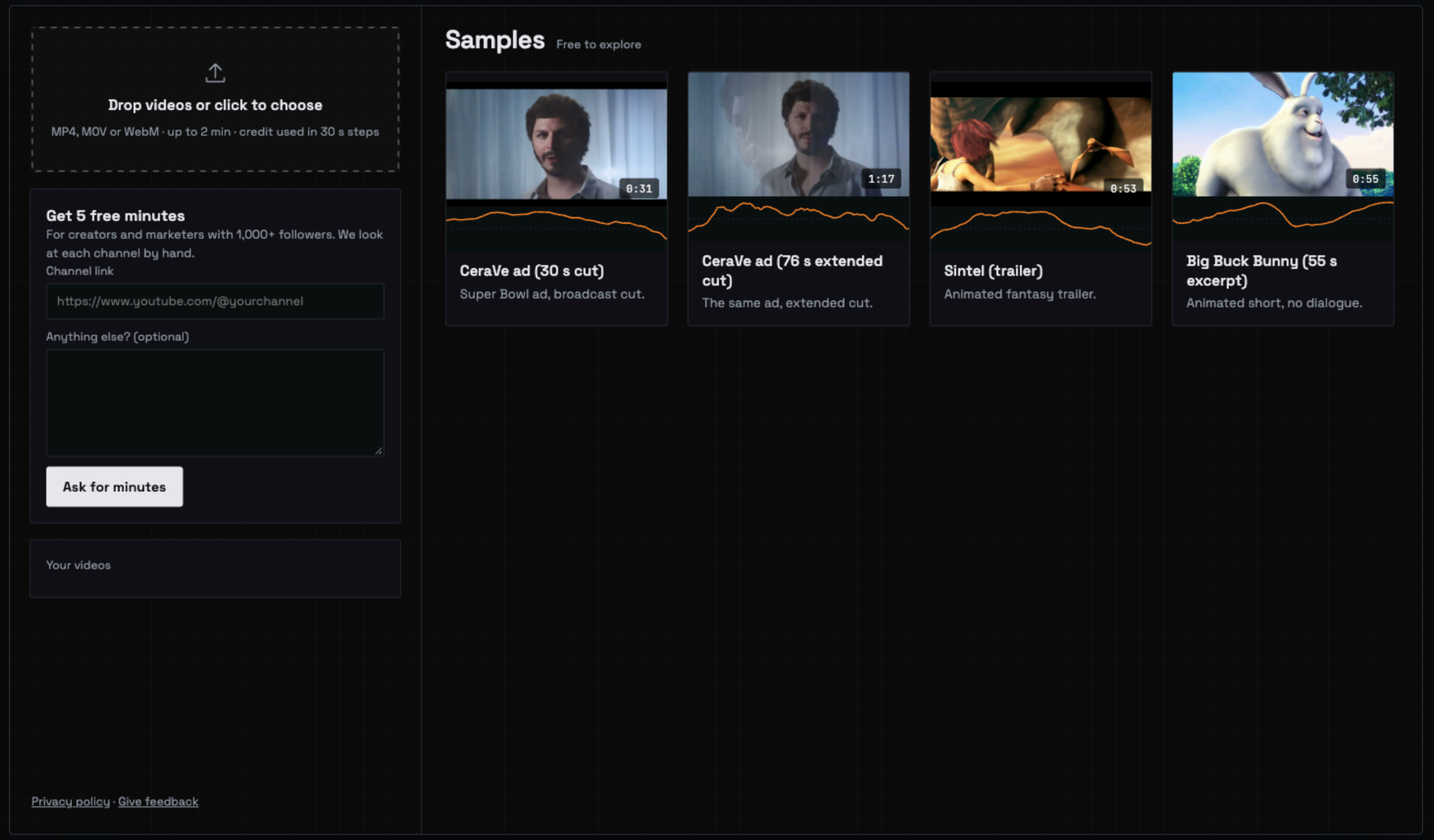
03
Predict
A GPU starts on demand

04
Watch
Live progress, stage by stage

05
Explore
Curve · 3D brain · report · compare

MP4, MOV, WEBM · UP TO 2 MINUTES · CREDIT USED IN 30 S STEPS

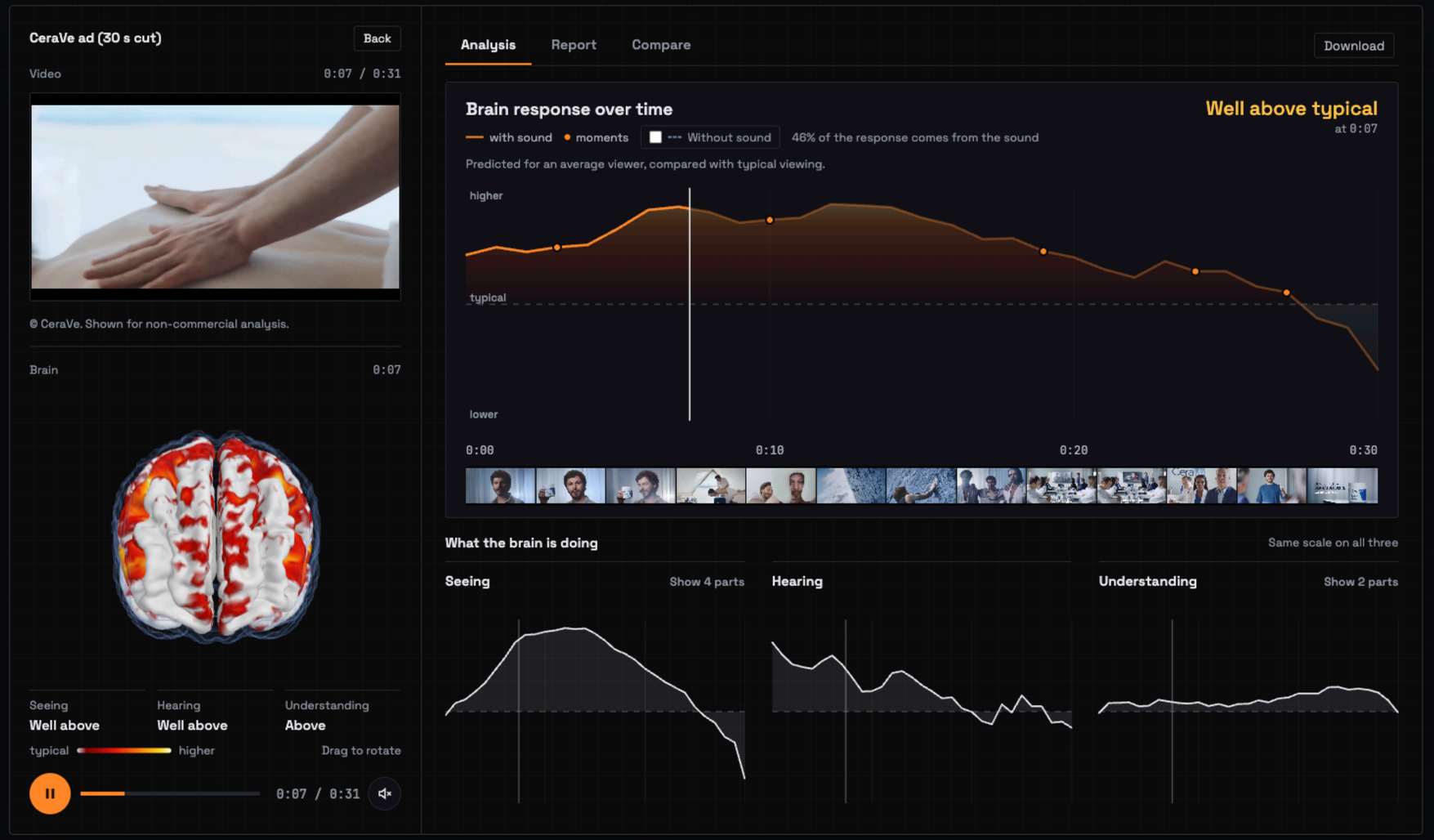
LIVE Sign in, upload, watch it process



1. Sign in with Google
2. The price shows before you confirm
3. Stages tick: check, queue, analyse

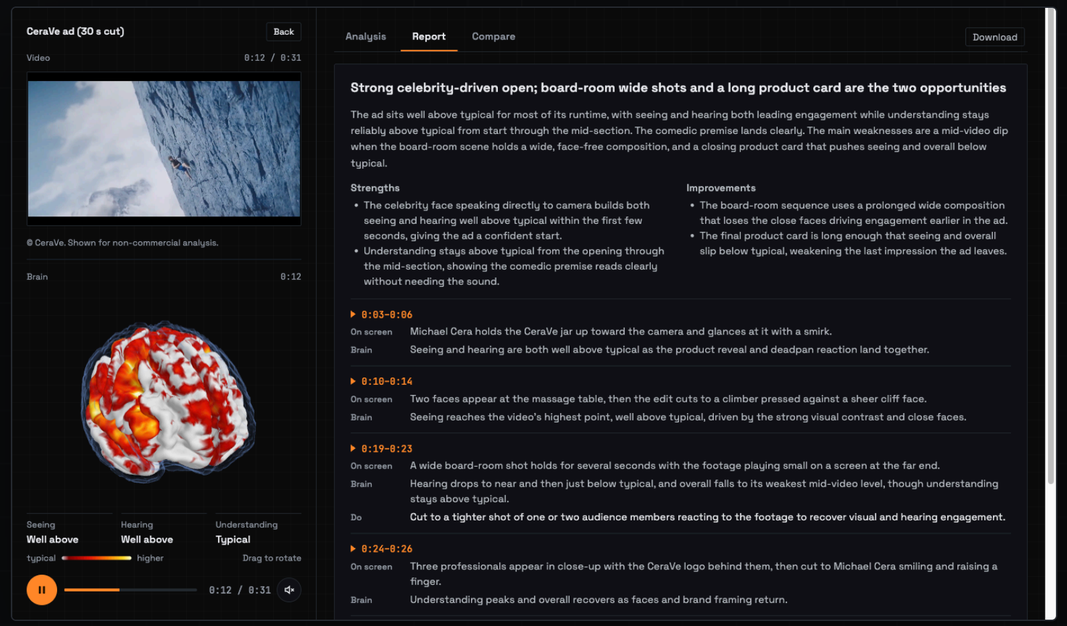
● LIVE

The results, second by second

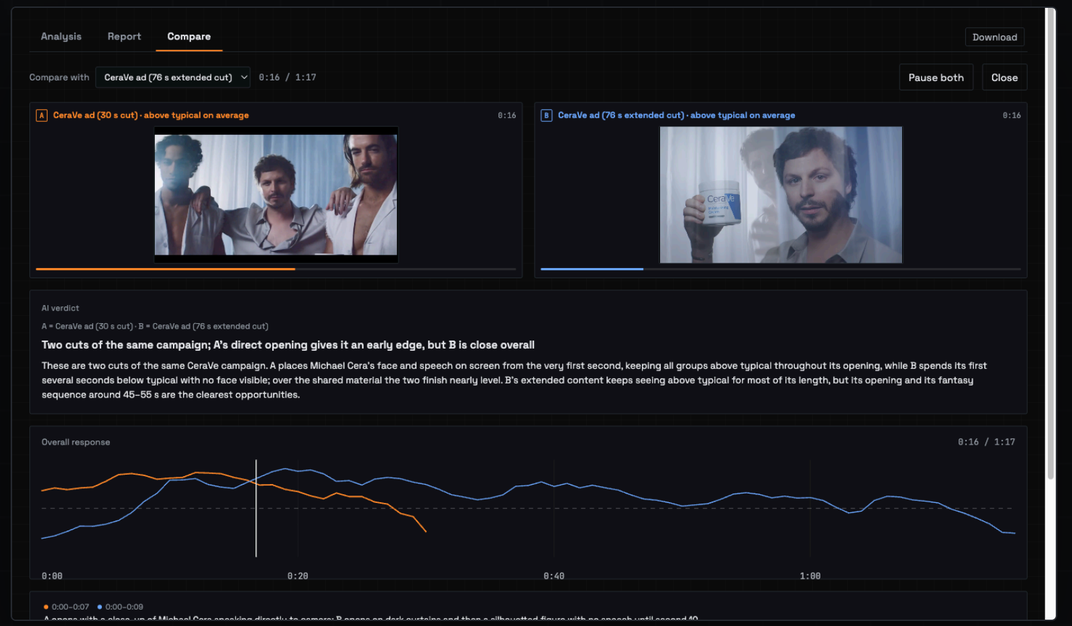


1. Follow the playhead
2. Peak near 0:06, dip at the end
3. Seeing · hearing · understanding
4. Toggle: without sound

LIVE The AI report, and comparing two cuts



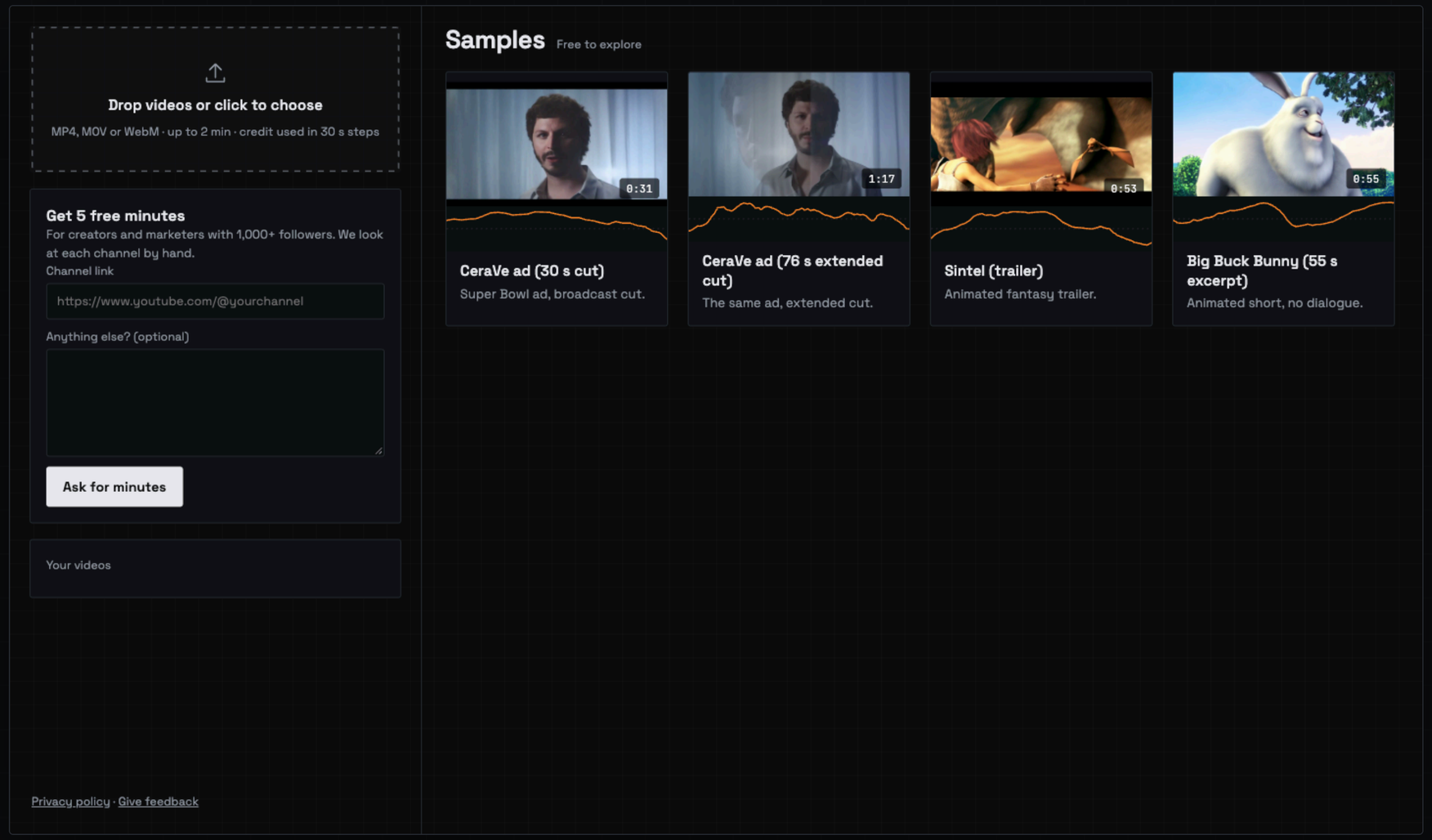
REPORT TAB



COMPARE TAB · 30 S CUT VS 76 S CUT

LIVE

A bad upload costs nothing



1. Upload a text file renamed .mp4
2. It is rejected within seconds
3. No credit is charged

REJECTED BEFORE ANY GPU STARTS

We pay for GPU minutes only when someone is waiting.



Check first

Size, length and price are verified before any GPU wakes.

Scale to zero

Off after 7 idle minutes. A guard stops a stuck GPU.

Refund by default

A failed job returns its credit automatically.

Free today. Paid tomorrow.

TODAY · RESEARCH

Free to use

The model is CC BY-NC, so we can’t sell it. Creators with 1,000+ followers get 5 free minutes.

NEXT · COMMERCIAL

Pay per analysis

Prepaid credits, per video-second, once we hold a commercial licence. A Stripe version is built and tested.

Creators share a retention curve → earn the analysis back → we get ground truth for the model.

What one analysis costs us.

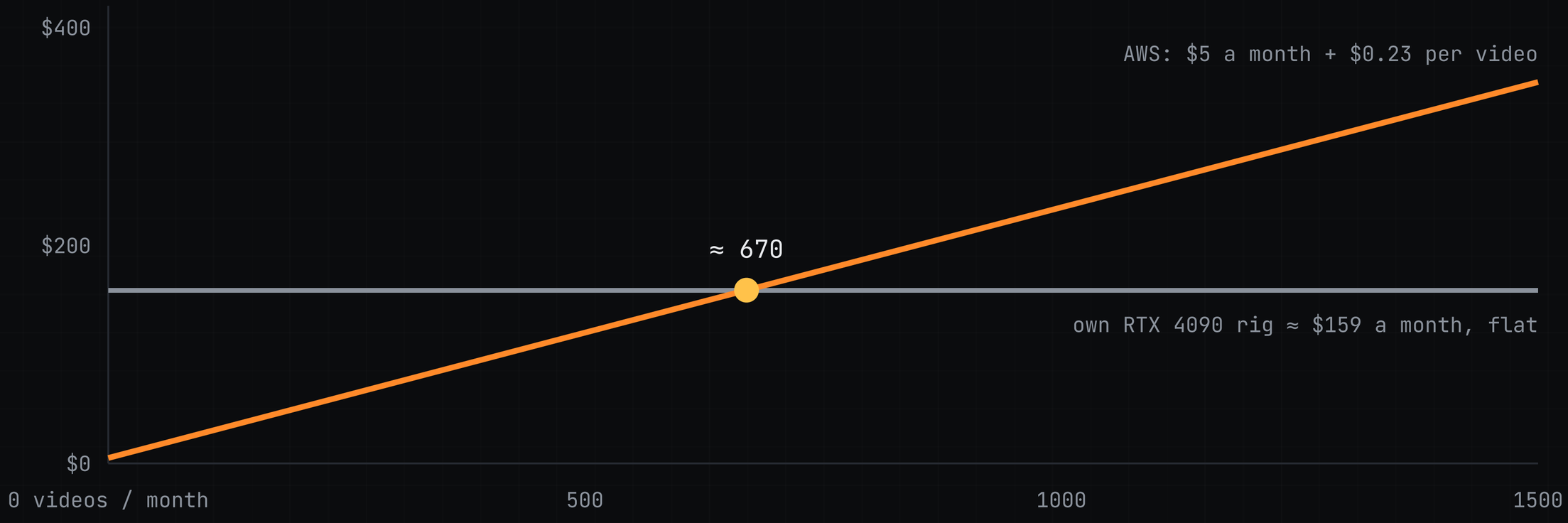
≈ $1

planned price per 30 s video

GPU time × price per hour	$0.13
Start-up + 7 idle minutes	$0.10
AI report (LLM)	[TBD]
Storage + database	$5 / month
Cost per 30 s video	**$0.23**
Gross margin	**≈ 77%**

393 s of GPU time at $1.22 an hour; 7.5 min start-up + 7 idle min shared by 3 videos. With 30 videos a session: $0.14 and ≈ 86%. AI report cost not included.

Cloud wins below about 670 videos a month.



What it claims, and what it doesn’t.

WHAT IT IS

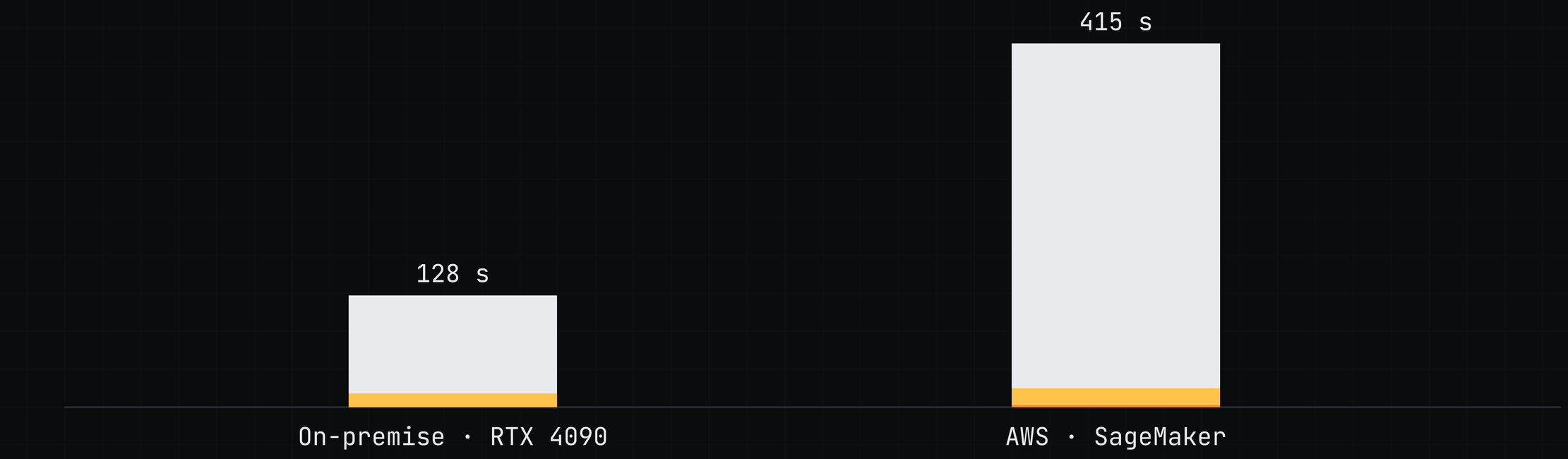
- A typical viewer’s predicted brain response
- One scale for every video
- A virtual pre-test for editing

WHAT IT ISN'T

- A prediction of views or sales
- Emotion classification
- Cleared for commercial use
- Statistically proven (n = 5)

Where does the time go in one job?

One 30 s clip, split into its stages, on our own GPU and on AWS.



GPU-SECONDS PER VIDEO-SECOND: 13.1 ON AWS (L4) · 3.71 ON AN RTX 4090 · COLD START: ≈ 7.5 MIN WAIT

Twenty people at once.

Bursts of 1, 5, 10 and 20 simultaneous uploads, with the GPU warm and cold.

RESULTS PENDING

1 user 5 users 10 users 20 users

GPU PRE-WARMED COLD START

BY DESIGN NO JOB IS DROPPED. A GPU FAILURE REFUNDS THE JOB AT ONCE.

Can people use it?

Five student creators. One task: upload an ad, find one peak and one drop-off.

[TBD] / 100

SUS SCORE · 68 IS AVERAGE · 80 IS AN A

COMPLETED UNAIDED, OF 5 · MEDIAN TIME ON TASK [TBD]

A $10,000 brain study, in minutes.

Live today: sign-in, upload, prediction, 3D brain, AI report, comparison, refunds.

Next

Licence

A commercial model licence unlocks the paid phase.

Next

Validate

Creators' retention curves against our predictions.

Next

Polish

Fix what the five users flagged.

Thank you.
Questions?

djlh10wxwg1q9.cloudfront.net

CODE · GITHUB.COM/KELVINLOWHM86/NEUROLENS

Recorded demo

[Embed the recorded demo of slides 8–11 here]